In this chapter we will develop the theory of differentiation. Integration theory, including the fundamental theorem that relates differentiation and integration, will be the subject of the next chapter. We will assume that the reader is already familiar with the geometrical and physical interpretations of the derivative of a function as described in introductory calculus courses. Consequently, we will concentrate on the mathematical aspects of the derivative and not go into its applications in geometry, physics, economics, and so on.

The first section is devoted to a presentation of the basic results concerning the differentiation of functions. In Section 6.2 we discuss the fundamental Mean Value Theorem and some of its applications. In Section 6.3 the important L'Hospital Rules are presented for the calculation of certain types of "indeterminate" limits.

In Section 6.4 we give a brief discussion of Taylor's Theorem and a few of its applications—for example, to convex functions and to Newton's Method for the location of roots.

Section 6.1 The Derivative

In this section we will present some of the elementary properties of the derivative. We begin with the definition of the derivative of a function.

6.1.1 Definition Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that a real number L is the **derivative of f at c** if given any $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that if $x \in I$ satisfies $0 < |x - c| < \delta(\varepsilon)$, then

$$\left|\frac{f(x) - f(c)}{x - c} - L\right| < \varepsilon. \tag{1}$$

In this case we say that f is **differentiable** at c, and we write $f'(c)$ for L.

In other words, the derivative of f at c is given by the limit

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \tag{2}$$

provided this limit exists. (We allow the possibility that c may be the endpoint of the interval.)

Note It is possible to define the derivative of a function having a domain more general than an interval (since the point c need only be an element of the domain and also a cluster point of the domain) but the significance of the concept is most naturally apparent for functions defined on intervals. Consequently we shall limit our attention to such functions.

Whenever the derivative of $f : I \to \mathbb{R}$ exists at a point $c \in I$, its value is denoted by $f'(c)$. In this way we obtain a function f' whose domain is a subset of the domain of f. In working with the function f', it is convenient to regard it also as a function of x. For example, if $f(x) := x^2$ for $x \in \mathbb{R}$, then at any c in $\mathbb{R}$ we have

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = \lim_{x \to c} \frac{x^2 - c^2}{x - c} = \lim_{x \to c} (x + c) = 2c.$$

Thus, in this case, the function f' is defined on all of $\mathbb{R}$ and $f'(x) = 2x$ for $x \in \mathbb{R}$.

We now show that continuity of f at a point c is a necessary (but not sufficient) condition for the existence of the derivative at c.

6.1.2 Theorem *If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then f is continuous at c.*

Proof. For all $x \in I$, $x \neq c$, we have

$$f(x) - f(c) = \left(\frac{f(x) - f(c)}{x - c}\right)(x - c).$$

Since $f'(c)$ exists, we may apply Theorem 4.2.4 concerning the limit of a product to conclude that

$$\begin{aligned}\lim_{x\to c}(f(x) - f(c)) &= \lim_{x\to c}\left(\frac{f(x) - f(c)}{x - c}\right)\left(\lim_{x\to c}(x - c)\right) \\ &= f'(c) \cdot 0 = 0.\end{aligned}$$

Therefore, $\lim_{x\to c} f(x) = f(c)$ so that f is continuous at c. Q.E.D.

The continuity of $f : I \to \mathbb{R}$ at a point does not assure the existence of the derivative at that point. For example, if $f(x) := |x|$ for $x \in \mathbb{R}$, then for $x \neq 0$ we have $(f(x) - f(0))/(x - 0) = |x|/x$, which is equal to 1 if $x > 0$, and equal to -1 if $x < 0$. Thus the limit at 0 does not exist [see Example 4.1.10(b)], and therefore the function is not differentiable at 0. Hence, continuity at a point c is *not* a sufficient condition for the derivative to exist at c.

Remark By taking simple algebraic combinations of functions of the form $x \mapsto |x - c|$, it is not difficult to construct continuous functions that do not have a derivative at a finite (or even a countable) number of points. In 1872, Karl Weierstrass astounded the mathematical world by giving an example of a function that is *continuous at every point but whose derivative does not exist anywhere.* Such a function defied geometric intuition about curves and tangent lines, and consequently spurred much deeper investigations into the concepts of real analysis. It can be shown that the function f defined by the series

$$f(x) := \sum_{n=0}^{\infty} \frac{1}{2^n}\cos(3^n x)$$

has the stated property. A very interesting historical discussion of this and other examples of continuous, nondifferentiable functions is given in Kline, pp. 955–966, and also in Hawkins, pp. 44–46. A detailed proof for a slightly different example can be found in Appendix E.

There are a number of basic properties of the derivative that are very useful in the calculation of the derivatives of various combinations of functions. We now provide the justification of some of these properties, which will be familiar to the reader from earlier courses.

6.1.3 Theorem *Let $I \subseteq \mathbb{R}$ be an interval, let $c \in I$, and let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be functions that are differentiable at c. Then:*

(a) *If $\alpha \in \mathbb{R}$, then the function αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c). \tag{3}$$

(b) *The function $f+g$ is differentiable at c, and*

$$(f+g)'(c) = f'(c) + g'(c). \tag{4}$$

(c) (*Product Rule*) *The function fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c). \tag{5}$$

(d) (*Quotient Rule*) *If $g(c) \neq 0$, then the function f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}. \tag{6}$$

Proof. We shall prove (c) and (d), leaving (a) and (b) as exercises for the reader.

(c) Let $p := fg$; then for $x \in I$, $x \neq c$, we have

$$\begin{aligned}\frac{p(x)-p(c)}{x-c} &= \frac{f(x)g(x) - f(c)g(c)}{x-c} \\ &= \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x-c} \\ &= \frac{f(x)-f(c)}{x-c}\cdot g(x) + f(c)\cdot\frac{g(x)-g(c)}{x-c}.\end{aligned}$$

Since g is continuous at c, by Theorem 6.1.2, then $\lim_{x\to c} g(x) = g(c)$. Since f and g are differentiable at c, we deduce from Theorem 4.2.4 on properties of limits that

$$\lim_{x\to c}\frac{p(x)-p(c)}{x-c} = f'(c)g(c) + f(c)g'(c).$$

Hence $p := fg$ is differentiable at c and (5) holds.

(d) Let $q := f/g$. Since g is differentiable at c, it is continuous at that point (by Theorem 6.1.2). Therefore, since $g(c) \neq 0$, we know from Theorem 4.2.9 that there exists an interval $J \subseteq I$ with $c \in J$ such that $g(x) \neq 0$ for all $x \in J$. For $x \in J$, $x \neq c$, we have

$$\begin{aligned}\frac{q(x)-q(c)}{x-c} &= \frac{f(x)/g(x) - f(c)/g(c)}{x-c} = \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)(x-c)} \\ &= \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)(x-c)} \\ &= \frac{1}{g(x)g(c)}\left[\frac{f(x)-f(c)}{x-c}\cdot g(c) - f(c)\cdot\frac{g(x)-g(c)}{x-c}\right].\end{aligned}$$

Using the continuity of g at c and the differentiability of f and g at c, we get

$$q'(c) = \lim_{x\to c}\frac{q(x)-q(c)}{x-c} = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Thus, $q = f/g$ is differentiable at c and equation (6) holds. Q.E.D.

Mathematical Induction may be used to obtain the following extensions of the differentiation rules.

6.1.4 Corollary *If $f_1, f_2, \ldots, f_n$ are functions on an interval I to $\mathbb{R}$ that are differentiable at $c \in I$, then:*

(a) *The function $f_1 + f_2 + \cdots + f_n$ is differentiable at c and*

$$(f_1 + f_2 + \cdots + f_n)'(c) = f_1'(c) + f_2'(c) + \cdots + f_n'(c). \tag{7}$$

(b) *The function $f_1 f_2 \cdots f_n$ is differentiable at c, and*

$$(f_1 f_2 \cdots f_n)'(c) = f_1'(c) f_2(c) \cdots f_n(c) + f_1(c) f_2'(c) \cdots f_n(c) + \cdots + f_1(c) f_2(c) \cdots f_n'(c). \tag{8}$$

An important special case of the extended product rule (8) occurs if the functions are equal, that is, $f_1 = f_2 = \cdots = f_n = f$. Then (8) becomes

$$(f^n)'(c) = n(f(c))^{n-1} f'(c). \tag{9}$$

In particular, if we take $f(x) := x$, then we find the derivative of $g(x) := x^n$ to be $g'(x) = nx^{n-1}$, $n \in \mathbb{N}$. The formula is extended to include negative integers by applying the Quotient Rule 6.1.3(d).

Notation If $I \subseteq \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$, we have introduced the notation f' to denote the function whose domain is a subset of I and whose value at a point c is the derivative $f'(c)$ of f at c. There are other notations that are sometimes used for f'; for example, one sometimes writes Df for f'. Thus one can write formulas (4) and (5) in the form:

$$D(f+g) = Df + Dg, \qquad D(fg) = (Df) \cdot g + f \cdot (Dg).$$

When x is the "independent variable," it is common practice in elementary courses to write df/dx for f'. Thus formula (5) is sometimes written in the form

$$\frac{d}{dx}(f(x)g(x)) = \left(\frac{df}{dx}(x)\right) g(x) + f(x) \left(\frac{dg}{dx}(x)\right).$$

This last notation, due to Leibniz, has certain advantages. However, it also has certain disadvantages and must be used with some care.

The Chain Rule

We now turn to the theorem on the differentiation of composite functions known as the "Chain Rule." It provides a formula for finding the derivative of a composite function $g \circ f$ in terms of the derivatives of g and f.

We first establish the following theorem concerning the derivative of a function at a point that gives us a very nice method for proving the Chain Rule. It will also be used to derive the formula for differentiating inverse functions.

6.1.5 Carathéodory's Theorem *Let f be defined on an interval I containing the point c. Then f is differentiable at c if and only if there exists a function φ on I that is continuous at c and satisfies*

$$f(x) - f(c) = \varphi(x)(x - c) \quad \text{for} \quad x \in I. \tag{10}$$

In this case, we have $\varphi(c) = f'(c)$.

Proof. ($\Rightarrow$) If $f'(c)$ exists, we can define φ by

$$\varphi(x) := \begin{cases} \dfrac{f(x) - f(c)}{x - c} & \text{for} \quad x \neq c,\ x \in I, \\ f'(c) & \text{for} \quad x = c. \end{cases}$$

The continuity of φ follows from the fact that $\lim_{x \to c} \varphi(x) = f'(c)$. If $x = c$, then both sides of (10) equal 0, while if $x \neq c$, then multiplication of $\varphi(x)$ by $x - c$ gives (10) for all other $x \in I$.

($\Leftarrow$) Now assume that a function φ that is continuous at c and satisfying (10) exists. If we divide (10) by $x - c \neq 0$, then the continuity of φ implies that

$$\varphi(c) = \lim_{x \to c} \varphi(x) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. Therefore f is differentiable at c and $f'(c) = \varphi(c)$. Q.E.D.

To illustrate Carathéodory's Theorem, we consider the function f defined by $f(x) := x^3$ for $x \in \mathbb{R}$. For $c \in \mathbb{R}$, we see from the factorization

$$x^3 - c^3 = (x^2 + cx + c^2)(x - c)$$

that $\varphi(x) := x^2 + cx + c^2$ satisfies the conditions of the theorem. Therefore, we conclude that f is differentiable at $c \in \mathbb{R}$ and that $f'(c) = \varphi(c) = 3c^2$.

We will now establish the Chain Rule. If f is differentiable at c and g is differentiable at $f(c)$, then the Chain Rule states that the derivative of the composite function $g \circ f$ at c is the product $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$. Note that this can be written as

$$(g \circ f)' = (g' \circ f) \cdot f'.$$

One approach to the Chain Rule is the observation that the difference quotient can be written, when $f(x) \neq f(c)$, as the product

$$\frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c}.$$

This suggests the correct limiting value. Unfortunately, the first factor in the product on the right is undefined if the denominator $f(x) - f(c)$ equals 0 for values of x near c, and this presents a problem. However, the use of Carathéodory's Theorem neatly avoids this difficulty.

6.1.6 Chain Rule *Let I, J be intervals in $\mathbb{R}$, let $g : I \to \mathbb{R}$ and $f : J \to \mathbb{R}$ be functions such that $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and if g is differentiable at $f(c)$, then the composite function $g \circ f$ is differentiable at c and*

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c). \tag{11}$$

Proof. Since $f'(c)$ exists, Carathéodory's Theorem 6.1.5 implies that there exists a function φ on J such that φ is continuous at c and $f(x) - f(c) = \varphi(x)(x - c)$ for $x \in J$, and where $\varphi(c) = f'(c)$. Also, since $g'(f(c))$ exists, there is a function ψ defined on I such that ψ is continuous at $d := f(c)$ and $g(y) - g(d) = \psi(y)(y - d)$ for $y \in I$, where $\psi(d) = g'(d)$. Substitution of $y = f(x)$ and $d = f(c)$ then produces

$$g(f(x)) - g(f(c)) = \psi(f(x))(f(x) - f(c)) = [(\psi \circ f(x)) \cdot \varphi(x)](x - c)$$

for all $x \in J$ such that $f(x) \in I$. Since the function $(\psi \circ f) \cdot \varphi$ is continuous at c and its value at c is $g'(f(c)) \cdot f'(c)$, Carathéodory's Theorem gives (11). Q.E.D.

If g is differentiable on I, if f is differentiable on J, and if $f(J) \subseteq I$, then it follows from the Chain Rule that $(g \circ f)' = (g' \circ f) \cdot f'$, which can also be written in the form $D(g \circ f) = (Dg \circ f) \cdot Df$.

6.1.7 Examples **(a)** If $f : I \to \mathbb{R}$ is differentiable on I and $g(y) := y^n$ for $y \in \mathbb{R}$ and $n \in \mathbb{N}$, then since $g'(y) = ny^{n-1}$, it follows from the Chain Rule 6.1.6 that

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) \quad \text{for} \quad x \in I.$$

Therefore we have $(f^n)'(x) = n(f(x))^{n-1}f'(x)$ for all $x \in I$ as was seen in (9).

(b) Suppose that $f : I \to \mathbb{R}$ is differentiable on I and that $f(x) \neq 0$ and $f'(x) \neq 0$ for $x \in I$. If $h(y) := 1/y$ for $y \neq 0$, then it is an exercise to show that $h'(y) = -1/y^2$ for $y \in \mathbb{R}$, $y \neq 0$. Therefore we have

$$\left(\frac{1}{f}\right)'(x) = (h \circ f)'(x) = h'(f(x))f'(x) = -\frac{f'(x)}{(f(x))^2} \quad \text{for} \quad x \in I.$$

(c) The absolute value function $g(x) := |x|$ is differentiable at all $x \neq 0$ and has derivative $g'(x) = \operatorname{sgn}(x)$ for $x \neq 0$. (The signum function is defined in Example 4.1.10(b).) Though sgn is defined everywhere, it is not equal to g' at $x = 0$ since $g'(0)$ does not exist.

Now if f is a differentiable function, then the Chain Rule implies that the function $g \circ f = |f|$ is also differentiable at all points x where $f(x) \neq 0$, and its derivative is given by

$$|f|'(x) = \operatorname{sgn}(f(x)) \cdot f'(x) = \begin{cases} f'(x) & \text{if} \quad f(x) > 0, \\ -f'(x) & \text{if} \quad f(x) < 0. \end{cases}$$

If f is differentiable at a point c with $f(c) = 0$, then it is an exercise to show that $|f|$ is differentiable at c if and only if $f'(c) = 0$. (See Exercise 7.)

For example, if $f(x) := x^2 - 1$ for $x \in \mathbb{R}$, then the derivative of its absolute value $|f|(x) = |x^2 - 1|$ is equal to $|f|'(x) = \operatorname{sgn}(x^2 - 1) \cdot (2x)$ for $x \neq 1, -1$. See Figure 6.1.1 for a graph of $|f|$.

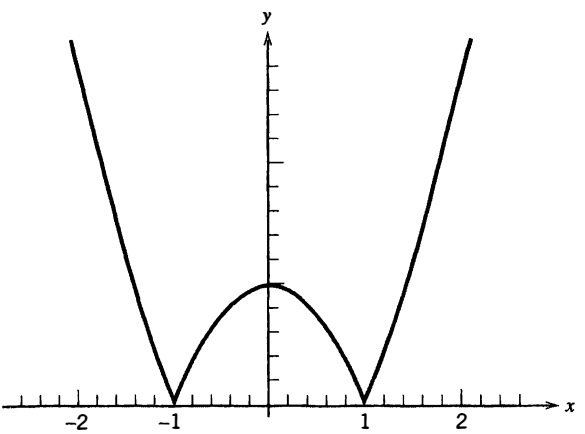


Figure 6.1.1 The function $|f|(x) = |x^2 - 1|$.

(d) It will be proved later that if $S(x) := \sin x$ and $C(x) := \cos x$ for all $x \in \mathbb{R}$, then

$$S'(x) = \cos x = C(x) \quad \text{and} \quad C'(x) = -\sin x = -S(x)$$

for all $x \in \mathbb{R}$. If we use these facts together with the definitions

$$\tan x := \frac{\sin x}{\cos x}, \quad \sec x := \frac{1}{\cos x},$$

for $x \neq (2k+1)\pi/2$, $k \in \mathbb{Z}$, and apply the Quotient Rule 6.1.3(d), we obtain

$$D\tan x = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2} = (\sec x)^2,$$

$$D\sec x = \frac{0 - 1(-\sin x)}{(\cos x)^2} = \frac{\sin x}{(\cos x)^2} = (\sec x)(\tan x)$$

for $x \neq (2k+1)\pi/2, k \in \mathbb{Z}$.

Similarly, since

$$\cot x := \frac{\cos x}{\sin x}, \qquad \csc x := \frac{1}{\sin x}$$

for $x \neq k\pi$, $k \in \mathbb{Z}$, then we obtain

$$D\cot x = -(\csc x)^2 \qquad \text{and} \qquad D\csc x = -(\csc x)(\cot x)$$

for $x \neq k\pi$, $k \in \mathbb{Z}$.

(e) Suppose that f is defined by

$$f(x) := \begin{cases} x^2 \sin(1/x) & \text{for } x \neq 0, \\ 0 & \text{for } x = 0. \end{cases}$$

If we use the fact that $D\sin x = \cos x$ for all $x \in \mathbb{R}$ and apply the Product Rule 6.1.3(c) and the Chain Rule 6.1.6, we obtain (why?)

$$f'(x) = 2x\sin(1/x) - \cos(1/x) \quad \text{for} \quad x \neq 0.$$

If $x = 0$, none of the calculational rules may be applied. (Why?) Consequently, the derivative of f at $x = 0$ must be found by applying the definition of derivative. We find that

$$f'(0) = \lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x\to 0} x\sin(1/x) = 0.$$

Hence, the derivative f' of f exists at all $x \in \mathbb{R}$. However, the function f' does not have a limit at $x = 0$ (why?), and consequently f' is discontinuous at $x = 0$. Thus, a function f that is differentiable at every point of $\mathbb{R}$ need not have a continuous derivative f'. □

Inverse Functions

We will now relate the derivative of a function to the derivative of its inverse function, when this inverse function exists. We will limit our attention to a continuous strictly monotone function and use the Continuous Inverse Theorem 5.6.5 to ensure the existence of a continuous inverse function.

If f is a continuous strictly monotone function on an interval I, then its inverse function $g = f^{-1}$ is defined on the interval $J := f(I)$ and satisfies the relation

$$g\big(f(x)\big) = x \quad \text{for} \quad x \in I.$$

If $c \in I$ and $d := f(c)$, and if we knew that both $f'(c)$ and $g'(d)$ exist, then we could differentiate both sides of the equation and apply the Chain Rule to the left side to get $g'\big(f(c)\big) \cdot f'(c) = 1$. Thus, if $f'(c) \neq 0$, we would obtain

$$g'(d) = \frac{1}{f'(c)}.$$

However, it is necessary to deduce the differentiability of the inverse function g from the assumed differentiability of f before such a calculation can be performed. This is nicely accomplished by using Carathéodory's Theorem.

6.1.8 Theorem *Let I be an interval in $\mathbb{R}$ and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Let $J := f(I)$ and let $g : J \to \mathbb{R}$ be the strictly monotone and continuous function inverse to f. If f is differentiable at $c \in I$ and $f'(c) \neq 0$, then g is differentiable at $d := f(c)$ and*

$$g'(d) = \frac{1}{f'(c)} = \frac{1}{f'(g(d))}. \tag{12}$$

Proof. Given $c \in \mathbb{R}$, we obtain from Carathéodory's Theorem 6.1.5 a function φ on I with properties that φ is continuous at c, $f(x) - f(c) = \varphi(x)(x - c)$ for $x \in I$, and $\varphi(c) = f'(c)$. Since $\varphi(c) \neq 0$ by hypothesis, there exists a neighborhood $V := (c - \delta,\ c + \delta)$ such that $\varphi(x) \neq 0$ for all $x \in V \cap I$. (See Theorem 4.2.9.) If $U := f(V \cap I)$, then the inverse function g satisfies $f(g(y)) = y$ for all $y \in U$, so that

$$y - d = f(g(y)) - f(c) = \varphi(g(y)) \cdot (g(y) - g(d)).$$

Since $\varphi(g(y)) \neq 0$ for $y \in U$, we can divide to get

$$g(y) - g(d) = \frac{1}{\varphi(g(y))} \cdot (y - d).$$

Since the function $1/(\varphi \circ g)$ is continuous at d, we apply Theorem 6.1.5 to conclude that $g'(d)$ exists and $g'(d) = 1/\varphi(g(d)) = 1/\varphi(c) = 1/f'(c)$. Q.E.D.

Note The hypothesis, made in Theorem 6.1.8, that $f'(c) \neq 0$ is essential. In fact, if $f'(c) = 0$, then the inverse function g is *never* differentiable at $d = f(c)$, since the assumed existence of $g'(d)$ would lead to $1 = f'(c)g'(d) = 0$, which is impossible. The function $f(x) := x^3$ with $c = 0$ is such an example.

6.1.9 Theorem *Let I be an interval and let $f : I \to \mathbb{R}$ be strictly monotone on I. Let $J := f(I)$ and let $g : J \to \mathbb{R}$ be the function inverse to f. If f is differentiable on I and $f'(x) \neq 0$ for $x \in I$, then g is differentiable on J and*

$$g' = \frac{1}{f' \circ g}. \tag{13}$$

Proof. If f is differentiable on I, then Theorem 6.1.2 implies that f is continuous on I, and by the Continuous Inverse Theorem 5.6.5, the inverse function g is continuous on J. Equation (13) now follows from Theorem 6.1.8. Q.E.D.

Remark If f and g are the functions of Theorem 6.1.9, and if $x \in I$ and $y \in J$ are related by $y = f(x)$ and $x = g(y)$, then equation (13) can be written in the form

$$g'(y) = \frac{1}{(f' \circ g)(y)}, \quad y \in J, \qquad \text{or} \qquad (g' \circ f)(x) = \frac{1}{f'(x)}, \quad x \in I.$$

It can also be written in the form $g'(y) = 1/f'(x)$, *provided* that it is kept in mind that x and y are related by $y = f(x)$ and $x = g(y)$.

6.1.10 Examples **(a)** The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := x^5 + 4x + 3$ is continuous and strictly monotone increasing (since it is the sum of two strictly increasing functions). Moreover, $f'(x) = 5x^4 + 4$ is never zero. Therefore, by Theorem 6.1.8, the inverse function $g = f^{-1}$ is differentiable at every point. If we take $c = 1$, then since $f(1) = 8$, we obtain $g'(8) = g'(f(1)) = 1/f'(1) = 1/9$.

(b) Let $n \in \mathbb{N}$ be even, let $I := [0, \infty)$, and let $f(x) := x^n$ for $x \in I$. It was seen at the end of Section 5.6 that f is strictly increasing and continuous on I, so that its inverse function $g(y) := y^{1/n}$ for $y \in J := [0, \infty)$ is also strictly increasing and continuous on J. Moreover, we have $f'(x) = nx^{n-1}$ for all $x \in I$. Hence it follows that if $y > 0$, then $g'(y)$ exists and

$$g'(y) = \frac{1}{f'(g(y))} = \frac{1}{n(g(y))^{n-1}} = \frac{1}{ny^{(n-1)/n}}.$$

Hence we deduce that

$$g'(y) = \frac{1}{n} y^{(1/n)-1} \quad \text{for} \quad y > 0.$$

However, g is *not* differentiable at 0. (For a graph of f and g, see Figures 5.6.4 and 5.6.5.)

(c) Let $n \in \mathbb{N}$, $n \neq 1$, be odd, let $F(x) := x^n$ *for* $x \in \mathbb{R}$, and let $G(y) := y^{1/n}$ be its inverse function defined for all $y \in \mathbb{R}$. As in part (b) we find that G is differentiable for $y \neq 0$ and that $G'(y) = (1/n)y^{(1/n)-1}$ for $y \neq 0$. However, G is not differentiable at 0, even though G is differentiable for all $y \neq 0$. (For a graph of F and G, see Figures 5.6.6 and 5.6.7.)

(d) Let $r := m/n$ be a positive rational number, let $I := [0, \infty)$, and let $R(x) := x^r$ for $x \in I$. (Recall Definition 5.6.6.) Then R is the composition of the functions $f(x) := x^m$ and $g(x) := x^{1/n}$, $x \in I$. That is, $R(x) = f(g(x))$ for $x \in I$. If we apply the Chain Rule 6.1.6 and the results of (b) [or (c), depending on whether n is even or odd], then we obtain

$$\begin{aligned} R'(x) &= f'(g(x))g'(x) = m\left(x^{1/n}\right)^{m-1} \cdot \frac{1}{n} x^{(1/n)-1} \\ &= \frac{m}{n} x^{(m/n)-1} = rx^{r-1} \end{aligned}$$

for all $x > 0$. If $r > 1$, then it is an exercise to show that the derivative also exists at $x = 0$ and $R'(0) = 0$. (For a graph of R see Figure 5.6.8.)

(e) The sine function is strictly increasing on the interval $I := [-\pi/2, \pi/2]$; therefore its inverse function, which we will denote by Arcsin, exists on $J := [-1, 1]$. That is, if $x \in [-\pi/2, \pi/2]$ and $y \in [-1, 1]$ then $y = \sin x$ if and only if $\text{Arcsin } y = x$. It was asserted (without proof) in Example 6.1.7(d) that sin is differentiable on I and that $D \sin x = \cos x$ for $x \in I$. Since $\cos x \neq 0$ for x in $(-\pi/2, \pi/2)$ it follows from Theorem 6.1.8 that

$$\begin{aligned} D \operatorname{Arcsin} y &= \frac{1}{D \sin x} = \frac{1}{\cos x} \\ &= \frac{1}{\sqrt{1 - (\sin x)^2}} = \frac{1}{\sqrt{1 - y^2}} \end{aligned}$$

for all $y \in (-1, 1)$. The derivative of Arcsin does *not* exist at the points -1 and 1. □

Exercises for Section 6.1

1. Use the definition to find the derivative of each of the following functions:
   (a) $f(x) := x^3$ for $x \in \mathbb{R}$,
   (b) $g(x) := 1/x$ for $x \in \mathbb{R}$, $x \neq 0$,
   (c) $h(x) := \sqrt{x}$ for $x > 0$,
   (d) $k(x) := 1/\sqrt{x}$ for $x > 0$.

2. Show that $f(x) := x^{1/3}$, $x \in \mathbb{R}$, is not differentiable at $x = 0$.
3. Prove Theorem 6.1.3(a), (b).
4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := x^2$ for x rational, $f(x) := 0$ for x irrational. Show that f is differentiable at $x = 0$, and find $f'(0)$.
5. Differentiate and simplify:
   (a) $f(x) := \dfrac{x}{1+x^2}$, (b) $g(x) := \sqrt{5-2x+x^2}$,
   (c) $h(x) := (\sin x^k)^m$ for $m, k \in \mathbb{N}$, (d) $k(x) := \tan(x^2)$ for $|x| < \sqrt{\pi/2}$.
6. Let $n \in \mathbb{N}$ and let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := x^n$ for $x \geq 0$ and $f(x) := 0$ for $x < 0$. For which values of n is f' continuous at 0? For which values of n is f' differentiable at 0?
7. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable at c and that $f(c) = 0$. Show that $g(x) := |f(x)|$ is differentiable at c if and only if $f'(c) = 0$.
8. Determine where each of the following functions from $\mathbb{R}$ to $\mathbb{R}$ is differentiable and find the derivative:
   (a) $f(x) := |x| + |x+1|$, (b) $g(x) := 2x + |x|$,
   (c) $h(x) := x|x|$, (d) $k(x) := |\sin x|$.
9. Prove that if $f : \mathbb{R} \to \mathbb{R}$ is an **even function** [that is, $f(-x) = f(x)$ for all $x \in \mathbb{R}$] and has a derivative at every point, then the derivative f' is an **odd function** [that is, $f'(-x) = -f'(x)$ for all $x \in \mathbb{R}$]. Also prove that if $g : \mathbb{R} \to \mathbb{R}$ is a differentiable odd function, then g' is an even function.
10. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) := x^2 \sin(1/x^2)$ for $x \neq 0$, and $g(0) := 0$. Show that g is differentiable for all $x \in \mathbb{R}$. Also show that the derivative g' is not bounded on the interval $[-1, 1]$.
11. Assume that there exists a function $L : (0, \infty) \to \mathbb{R}$ such that $L'(x) = 1/x$ for $x > 0$. Calculate the derivatives of the following functions:
   (a) $f(x) := L(2x+3)$ for $x > 0$, (b) $g(x) := (L(x^2))^3$ for $x > 0$,
   (c) $h(x) := L(ax)$ for $a > 0$, $x > 0$, (d) $k(x) := L(L(x))$ when $L(x) > 0$, $x > 0$.
12. If $r > 0$ is a rational number, let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := x^r \sin(1/x)$ for $x \neq 0$, and $f(0) := 0$. Determine those values of r for which $f'(0)$ exists.
13. If $f : \mathbb{R} \to \mathbb{R}$ is differentiable at $c \in \mathbb{R}$, show that
$$f'(c) = \lim(n\{f(c + 1/n) - f(c)\}).$$
However, show by example that the existence of the limit of this sequence does not imply the existence of $f'(c)$.
14. Given that the function $h(x) := x^3 + 2x + 1$ for $x \in \mathbb{R}$ has an inverse h^{-1} on $\mathbb{R}$, find the value of $(h^{-1})'(y)$ at the points corresponding to $x = 0, 1, -1$.
15. Given that the restriction of the cosine function cos to $I := [0, \pi]$ is strictly decreasing and that $\cos 0 = 1$, $\cos \pi = -1$, let $J := [-1, 1]$, and let Arccos: $J \to \mathbb{R}$ be the function inverse to the restriction of cos to I. Show that Arccos is differentiable on $(-1, 1)$ and $D \operatorname{Arccos} y = (-1)/(1-y^2)^{1/2}$ for $y \in (-1, 1)$. Show that Arccos is not differentiable at -1 and 1.
16. Given that the restriction of the tangent function tan to $I := (-\pi/2, \pi/2)$ is strictly increasing and that $\tan(I) = \mathbb{R}$, let Arctan: $\mathbb{R} \to \mathbb{R}$ be the function inverse to the restriction of tan to I. Show that Arctan is differentiable on $\mathbb{R}$ and that $D \operatorname{Arctan}(y) = (1+y^2)^{-1}$ for $y \in \mathbb{R}$.
17. Let $f : I \to \mathbb{R}$ be differentiable at $c \in I$. Establish the **Straddle Lemma:** Given $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that if $u, v \in I$ satisfy $c - \delta(\varepsilon) < u \leq c \leq v < c + \delta(\varepsilon)$, then we have $|f(v) - f(u) - (v-u)f'(c)| \leq \varepsilon(v-u)$. [*Hint*: The $\delta(\varepsilon)$ is given by Definition 6.1.1. Subtract and add the term $f(c) - cf'(c)$ on the left side and use the Triangle Inequality.]

Section 6.2 The Mean Value Theorem

The Mean Value Theorem, which relates the values of a function to values of its derivative, is one of the most useful results in real analysis. In this section we will establish this important theorem and sample some of its many consequences.

We begin by looking at the relationship between the relative extrema of a function and the values of its derivative. Recall that the function $f : I \to \mathbb{R}$ is said to have a **relative maximum** [respectively, **relative minimum**] at $c \in I$ if there exists a neighborhood $V := V_\delta(c)$ of c such that $f(x) \leq f(c)$ [respectively, $f(c) \leq f(x)$] for all x in $V \cap I$. We say that f has a **relative extremum** at $c \in I$ if it has either a relative maximum or a relative minimum at c.

The next result provides the theoretical justification for the familiar process of finding points at which f has relative extrema by examining the zeros of the derivative. However, it must be realized that this procedure applies only to *interior* points of the interval. For example, if $f(x) := x$ on the interval $I := [0, 1]$, then the endpoint $x = 0$ yields the unique relative minimum and the endpoint $x = 1$ yields the unique maximum of f on I, but neither point is a zero of the derivative of f.

6.2.1 Interior Extremum Theorem *Let c be an interior point of the interval I at which $f : I \to \mathbb{R}$ has a relative extremum. If the derivative of f at c exists, then $f'(c) = 0$.*

Proof. We will prove the result only for the case that f has a relative maximum at c; the proof for the case of a relative minimum is similar.

If $f'(c) > 0$, then by Theorem 4.2.9 there exists a neighborhood $V \subseteq I$ of c such that

$$\frac{f(x) - f(c)}{x - c} > 0 \quad \text{for} \quad x \in V,\ x \neq c.$$

If $x \in V$ and $x > c$, then we have

$$f(x) - f(c) = (x - c) \cdot \frac{f(x) - f(c)}{x - c} > 0.$$

But this contradicts the hypothesis that f has a relative maximum at c. Thus we cannot have $f'(c) > 0$. Similarly (how?), we cannot have $f'(c) < 0$. Therefore we must have $f'(c) = 0$. Q.E.D.

6.2.2 Corollary *Let $f : I \to \mathbb{R}$ be continuous on an interval I and suppose that f has a relative extremum at an interior point c of I. Then either the derivative of f at c does not exist, or it is equal to zero.*

We note that if $f(x) := |x|$ on $I := [-1, 1]$, then f has an interior minimum at $x = 0$; however, the derivative of f fails to exist at $x = 0$.

6.2.3 Rolle's Theorem *Suppose that f is continuous on a closed interval $I := [a, b]$, that the derivative f' exists at every point of the open interval (a, b), and that $f(a) = f(b) = 0$. Then there exists at least one point c in (a, b) such that $f'(c) = 0$.*

Proof. If f vanishes identically on I, then any c in (a, b) will satisfy the conclusion of the theorem. Hence we suppose that f does not vanish identically; replacing f by $-f$ if necessary, we may suppose that f assumes some positive values. By the Maximum-Minimum Theorem 5.3.4, the function f attains the value $\sup\{f(x) : x \in I\} > 0$ at some point c in I. Since $f(a) = f(b) = 0$, the point c must lie in (a, b); therefore $f'(c)$ exists.

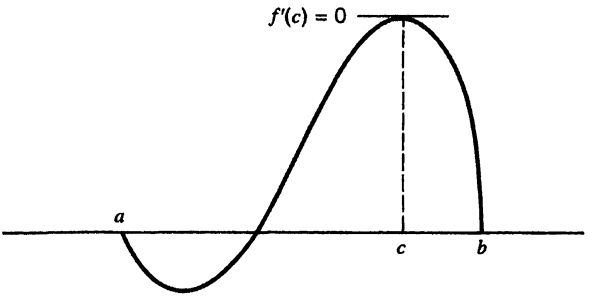


Figure 6.2.1 Rolle's Theorem

Since f has a relative maximum at c, we conclude from the Interior Extremum Theorem 6.2.1 that $f'(c) = 0$. (See Figure 6.2.1.) Q.E.D.

As a consequence of Rolle's Theorem, we obtain the fundamental Mean Value Theorem.

6.2.4 Mean Value Theorem *Suppose that f is continuous on a closed interval $I := [a, b]$, and that f has a derivative in the open interval (a, b). Then there exists at least one point c in (a, b) such that*

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. Consider the function φ defined on I by

$$\varphi(x) := f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

[The function φ is simply the difference of f and the function whose graph is the line segment joining the points $(a, f(a))$ and $(b, f(b))$; see Figure 6.2.2.] The hypotheses of

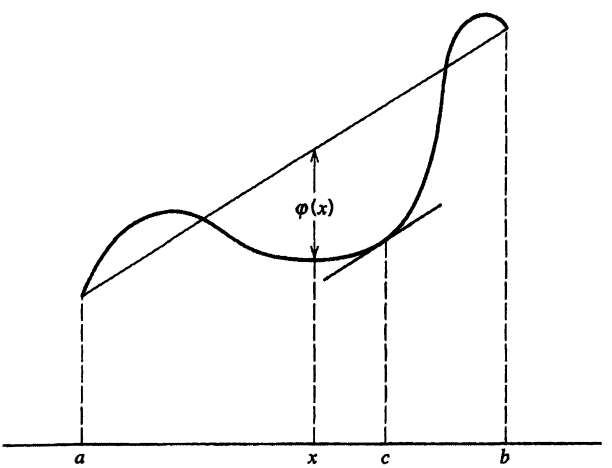


Figure 6.2.2 The Mean Value Theorem

Rolle's Theorem are satisfied by φ since φ is continuous on $[a, b]$, differentiable on (a, b), and $\varphi(a) = \varphi(b) = 0$. Therefore, there exists a point c in (a, b) such that

$$0 = \varphi'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

Hence, $f(b) - f(a) = f'(c)(b - a)$. Q.E.D.

Remark The geometric view of the Mean Value Theorem is that there is some point on the curve $y = f(x)$ at which the tangent line is parallel to the line segment through the points $(a, f(a))$ and $(b, f(b))$. Thus it is easy to remember the statement of the Mean Value Theorem by drawing appropriate diagrams. While this should not be discouraged, it tends to suggest that its importance is geometrical in nature, which is quite misleading. In fact the Mean Value Theorem is a wolf in sheep's clothing and is *the* Fundamental Theorem of Differential Calculus. In the remainder of this section, we will present some of the consequences of this result. Other applications will be given later.

The Mean Value Theorem permits one to draw conclusions about the nature of a function f from information about its derivative f'. The following results are obtained in this manner.

6.2.5 Theorem *Suppose that f is continuous on the closed interval $I := [a, b]$, that f is differentiable on the open interval (a, b), and that $f'(x) = 0$ for $x \in (a, b)$. Then f is constant on I.*

Proof. We will show that $f(x) = f(a)$ for all $x \in I$. Indeed, if $x \in I$, $x > a$, is given, we apply the Mean Value Theorem to f on the closed interval $[a, x]$. We obtain a point c (depending on x) between a and x such that $f(x) - f(a) = f'(c)(x - a)$. Since $f'(c) = 0$ (by hypothesis), we deduce that $f(x) - f(a) = 0$. Hence, $f(x) = f(a)$ for any $x \in I$. Q.E.D.

6.2.6 Corollary *Suppose that f and g are continuous on $I := [a, b]$, that they are differentiable on (a, b), and that $f'(x) = g'(x)$ for all $x \in (a, b)$. Then there exists a constant C such that $f = g + C$ on I.*

Recall that a function $f : I \to \mathbb{R}$ is said to be **increasing** on the interval I if whenever x_1, x_2 in I satisfy $x_1 < x_2$, then $f(x_1) \leq f(x_2)$. Also recall that f is **decreasing** on I if the function $-f$ is increasing on I.

6.2.7 Theorem *Let $f : I \to \mathbb{R}$ be differentiable on the interval I. Then:*

(a) *f is increasing on I if and only if $f'(x) \geq 0$ for all $x \in I$.*

(b) *f is decreasing on I if and only if $f'(x) \leq 0$ for all $x \in I$.*

Proof. (a) Suppose that $f'(x) \geq 0$ for all $x \in I$. If x_1, x_2 in I satisfy $x_1 < x_2$, then we apply the Mean Value Theorem to f on the closed interval $J := [x_1, x_2]$ to obtain a point c in (x_1, x_2) such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

Since $f'(c) \geq 0$ and $x_2 - x_1 > 0$, it follows that $f(x_2) - f(x_1) \geq 0$. (Why?) Hence, $f(x_1) \leq f(x_2)$ and, since $x_1 < x_2$ are arbitrary points in I, we conclude that f is increasing on I.

For the converse assertion, we suppose that f is differentiable and increasing on I. Thus, for any point $x \neq c$ in I, we have $(f(x) - f(c))/(x - c) \geq 0$. (Why?) Hence, by Theorem 4.2.6 we conclude that

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \geq 0.$$

(b) The proof of part (b) is similar and will be omitted. Q.E.D.

A function f is said to be **strictly increasing** on an interval I if for any points x_1, x_2 in I such that $x_1 < x_2$, we have $f(x_1) < f(x_2)$. An argument along the same lines of the proof of Theorem 6.2.7 can be made to show that a function having a strictly positive derivative on an interval is strictly increasing there. (See Exercise 13.) However, the converse assertion is not true, since a strictly increasing differentiable function may have a derivative that vanishes at certain points. For example, the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := x^3$ is strictly increasing on $\mathbb{R}$, but $f'(0) = 0$. The situation for strictly decreasing functions is similar.

Remark It is reasonable to define a function to be **increasing at a point** if there is a neighborhood of the point on which the function is increasing. One might suppose that, if the derivative is strictly positive at a point, then the function is increasing at this point. However, this supposition is false; indeed, the differentiable function defined by

$$g(x) := \begin{cases} x + 2x^2 \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

is such that $g'(0) = 1$, yet it can be shown that g is not increasing in any neighborhood of $x = 0$. (See Exercise 10.)

We next obtain a sufficient condition for a function to have a relative extremum at an interior point of an interval.

6.2.8 First Derivative Test for Extrema *Let f be continuous on the interval $I := [a, b]$ and let c be an interior point of I. Assume that f is differentiable on (a, c) and (c, b). Then:*

(a) *If there is a neighborhood $(c - \delta, c + \delta) \subseteq I$ such that $f'(x) \geq 0$ for $c - \delta < x < c$ and $f'(x) \leq 0$ for $c < x < c + \delta$, then f has a relative maximum at c.*

(b) *If there is a neighborhood $(c - \delta, c + \delta) \subseteq I$ such that $f'(x) \leq 0$ for $c - \delta < x < c$ and $f'(x) \geq 0$ for $c < x < c + \delta$, then f has a relative minimum at c.*

Proof. (a) If $x \in (c - \delta, c)$, then it follows from the Mean Value Theorem that there exists a point $c_x \in (x, c)$ such that $f(c) - f(x) = (c - x)f'(c_x)$. Since $f'(c_x) \geq 0$ we infer that $f(x) \leq f(c)$ for $x \in (c - \delta, c)$. Similarly, it follows (how?) that $f(x) \leq f(c)$ for $x \in (c, c + \delta)$. Therefore $f(x) \leq f(c)$ for all $x \in (c - \delta, c + \delta)$ so that f has a relative maximum at c.

(b) The proof is similar. Q.E.D.

Remark The converse of the First Derivative Test 6.2.8 is *not* true. For example, there exists a differentiable function $f : \mathbb{R} \to \mathbb{R}$ with absolute minimum at $x = 0$ but such that

f' takes on both positive and negative values on both sides of (and arbitrarily close to) $x = 0$. (See Exercise 9.)

Further Applications of the Mean Value Theorem

We will continue giving other types of applications of the Mean Value Theorem; in doing so we will draw more freely than before on the past experience of the reader and his or her knowledge concerning the derivatives of certain well-known functions.

6.2.9 Examples **(a)** Rolle's Theorem can be used for the location of roots of a function. For, if a function g can be identified as the derivative of a function f, then between any two roots of f there is at least one root of g. For example, let $g(x) := \cos x$, then g is known to be the derivative of $f(x) := \sin x$. Hence, between any two roots of $\sin x$ there is at least one root of $\cos x$. On the other hand, $g'(x) = -\sin x = -f(x)$, so another application of Rolle's Theorem tells us that between any two roots of cos there is at least one root of sin. Therefore, we conclude that the roots of sin and cos *interlace each other*. This conclusion is probably not news to the reader; however, the same type of argument can be applied to the *Bessel functions* J_n of order $n = 0, 1, 2, \ldots$ by using the relations

$$[x^n J_n(x)]' = x^n J_{n-1}(x), \quad [x^{n-1} J_n(x)]' = -x^{-n} J_{n+1}(x) \qquad \text{for} \quad x > 0.$$

The details of this argument should be supplied by the reader.

(b) We can apply the Mean Value Theorem for approximate calculations and to obtain error estimates. For example, suppose it is desired to evaluate $\sqrt{105}$. We employ the Mean Value Theorem with $f(x) := \sqrt{x}$, $a = 100$, $b = 105$, to obtain

$$\sqrt{105} - \sqrt{100} = \frac{5}{2\sqrt{c}}$$

for some number c with $100 < c < 105$. Since $10 < \sqrt{c} < \sqrt{105} < \sqrt{121} = 11$, we can assert that

$$\frac{5}{2(11)} < \sqrt{105} - 10 < \frac{5}{2(10)},$$

whence it follows that $10.2272 < \sqrt{105} < 10.2500$. This estimate may not be as sharp as desired. It is clear that the estimate $\sqrt{c} < \sqrt{105} < \sqrt{121}$ was wasteful and can be improved by making use of our conclusion that $\sqrt{105} < 10.2500$. Thus, $\sqrt{c} < 10.2500$ and we easily determine that

$$0.2439 < \frac{5}{2(10.2500)} < \sqrt{105} - 10.$$

Our improved estimate is $10.2439 < \sqrt{105} < 10.2500$. □

Inequalities

One very important use of the Mean Value Theorem is to obtain certain inequalities. Whenever information concerning the range of the derivative of a function is available, this information can be used to deduce certain properties of the function itself. The following examples illustrate the valuable role that the Mean Value Theorem plays in this respect.

6.2.10 Examples **(a)** The exponential function $f(x) := e^x$ has the derivative $f'(x) = e^x$ for all $x \in \mathbb{R}$. Thus $f'(x) > 1$ for $x > 0$, and $f'(x) < 1$ for $x < 0$. From these relationships, we will derive the inequality

$$e^x \geq 1 + x \quad \text{for} \quad x \in \mathbb{R}, \tag{1}$$

with equality occurring if and only if $x = 0$.

If $x = 0$, we have equality with both sides equal to 1. If $x > 0$, we apply the Mean Value Theorem to the function f on the interval $[0, x]$. Then for some c with $0 < c < x$ we have

$$e^x - e^0 = e^c(x - 0).$$

Since $e^0 = 1$ and $e^c > 1$, this becomes $e^x - 1 > x$ so that we have $e^x > 1 + x$ for $x > 0$. A similar argument establishes the same strict inequality for $x < 0$. Thus the inequality (1) holds for all x, and equality occurs only if $x = 0$.

(b) The function $g(x) := \sin x$ has the derivative $g'(x) = \cos x$ for all $x \in \mathbb{R}$. On the basis of the fact that $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$, we will show that

$$-x \leq \sin x \leq x \quad \text{for all} \quad x \geq 0. \tag{2}$$

Indeed, if we apply the Mean Value Theorem to g on the interval $[0, x]$, where $x > 0$, we obtain

$$\sin x - \sin 0 = (\cos c)(x - 0)$$

for some c between 0 and x. Since $\sin 0 = 0$ and $-1 \leq \cos c \leq 1$, we have $-x \leq \sin x \leq x$. Since equality holds at $x = 0$, the inequality (2) is established.

(c) (Bernoulli's inequality) If $\alpha > 1$, then

$$(1 + x)^\alpha \geq 1 + \alpha x \quad \text{for all} \quad x > -1, \tag{3}$$

with equality if and only if $x = 0$.

This inequality was established earlier, in Example 2.1.13(c), for positive integer values of α by using Mathematical Induction. We now derive the more general version by employing the Mean Value Theorem.

If $h(x) := (1 + x)^\alpha$ then $h'(x) = \alpha(1 + x)^{\alpha - 1}$ for all $x > -1$. [For rational α this derivative was established in Example 6.1.10(c). The extension to irrational will be discussed in Section 8.3.] If $x > 0$, we infer from the Mean Value Theorem applied to h on the interval $[0, x]$ that there exists c with $0 < c < x$ such that $h(x) - h(0) = h'(c)(x - 0)$. Thus, we have

$$(1 + x)^\alpha - 1 = \alpha(1 + c)^{\alpha - 1}x.$$

Since $c > 0$ and $\alpha - 1 > 0$, it follows that $(1 + c)^{\alpha - 1} > 1$ and hence that $(1 + x)^\alpha > 1 + \alpha x$. If $-1 < x < 0$, a similar use of the Mean Value Theorem on the interval $[x, 0]$ leads to the same strict inequality. Since the case $x = 0$ results in equality, we conclude that (3) is valid for all $x > -1$ with equality if and only if $x = 0$.

(d) Let α be a real number satisfying $0 < \alpha < 1$ and let $g(x) = \alpha x - x^\alpha$ for $x \geq 0$. Then $g'(x) = \alpha(1 - x^{\alpha - 1})$, so that $g'(x) < 0$ for $0 < x < 1$ and $g'(x) > 0$ for $x > 1$. Consequently, if $x \geq 0$, then $g(x) \geq g(1)$ and $g(x) = g(1)$ if and only if $x = 1$. Therefore, if $x \geq 0$ and $0 < \alpha < 1$, then we have

$$x^\alpha \leq \alpha x + (1 - \alpha).$$

If $a > 0$ and $b > 0$ and if we let $x = a/b$ and multiply by b, we obtain the inequality

$$a^{\alpha}b^{1-\alpha} \leq \alpha a + (1-\alpha)b,$$

where equality holds if and only if $a = b$. □

The Intermediate Value Property of Derivatives

We conclude this section with an interesting result, often referred to as Darboux's Theorem. It states that if a function f is differentiable at every point of an interval I, then the function f' has the Intermediate Value Property. This means that if f' takes on values A and B, then it also takes on all values between A and B. The reader will recognize this property as one of the important consequences of continuity as established in Theorem 5.3.7. It is remarkable that derivatives, which need not be continuous functions, also possess this property.

6.2.11 Lemma *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and assume that f has a derivative at c. Then:*

(a) *If $f'(c) > 0$, then there is a number $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ such that $c < x < c + \delta$.*

(b) *If $f'(c) < 0$, then there is a number $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ such that $c - \delta < x < c$.*

Proof. (a) Since

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c) > 0,$$

it follows from Theorem 4.2.9 that there is a number $\delta > 0$ such that if $x \in I$ and $0 < |x - c| < \delta$, then

$$\frac{f(x) - f(c)}{x - c} > 0.$$

If $x \in I$ also satisfies $x > c$, then we have

$$f(x) - f(c) = (x - c) \cdot \frac{f(x) - f(c)}{x - c} > 0.$$

Hence, if $x \in I$ and $c < x < c + \delta$, then $f(x) > f(c)$.

The proof of (b) is similar. Q.E.D.

6.2.12 Darboux's Theorem *If f is differentiable on $I = [a, b]$ and if k is a number between $f'(a)$ and $f'(b)$, then there is at least one point c in (a, b) such that $f'(c) = k$.*

Proof. Suppose that $f'(a) < k < f'(b)$. We define g on I by $g(x) := kx - f(x)$ for $x \in I$. Since g is continuous, it attains a maximum value on I. Since $g'(a) = k - f'(a) > 0$, it follows from Lemma 6.2.11(a) that the maximum of g does not occur at $x = a$. Similarly, since $g'(b) = k - f'(b) < 0$, it follows from Lemma 6.2.11(b) that the maximum does not occur at $x = b$. Therefore, g attains its maximum at some c in (a, b). Then from Theorem 6.2.1 we have $0 = g'(c) = k - f'(c)$. Hence, $f'(c) = k$. Q.E.D.

6.2.13 Example The function g: $[-1, 1] \to \mathbb{R}$ defined by

$$g(x) := \begin{cases} 1 & \text{for } 0 < x \leq 1, \\ 0 & \text{for } x = 0, \\ -1 & \text{for } -1 \leq x < 0, \end{cases}$$

(which is a restriction of the signum function) clearly fails to satisfy the intermediate value property on the interval $[-1, 1]$. Therefore, by Darboux's Theorem, there does not exist a function f such that $f'(x) = g(x)$ for all $x \in [-1, 1]$. In other words, g is *not* the derivative on $[-1, 1]$ of any function. □

Exercises for Section 6.2

1. For each of the following functions on $\mathbb{R}$ to $\mathbb{R}$, find points of relative extrema, the intervals on which the function is increasing, and those on which it is decreasing:
   (a) $f(x) := x^2 - 3x + 5$, (b) $g(x) := 3x - 4x^2$,
   (c) $h(x) := x^3 - 3x - 4$, (d) $k(x) := x^4 + 2x^2 - 4$.

2. Find the points of relative extrema, the intervals on which the following functions are increasing, and those on which they are decreasing:
   (a) $f(x) := x + 1/x$ for $x \neq 0$, (b) $g(x) := x/(x^2 + 1)$ for $x \in \mathbb{R}$,
   (c) $h(x) := \sqrt{x} - 2\sqrt{x+2}$ for $x > 0$, (d) $k(x) := 2x + 1/x^2$ for $x \neq 0$.

3. Find the points of relative extrema of the following functions on the specified domain:
   (a) $f(x) := |x^2 - 1|$ for $-4 \leq x \leq 4$, (b) $g(x) := 1 - (x-1)^{2/3}$ for $0 \leq x \leq 2$,
   (c) $h(x) := x|x^2 - 12|$ for $-2 \leq x \leq 3$, (d) $k(x) := x(x-8)^{1/3}$ for $0 \leq x \leq 9$.

4. Let $a_1, a_2, \ldots, a_n$ be real numbers and let f be defined on $\mathbb{R}$ by
$$f(x) := \sum_{i=1}^{n} (a_i - x)^2 \quad \text{for} \quad x \in \mathbb{R}.$$
   Find the unique point of relative minimum for f.

5. Let $a > b > 0$ and let $n \in \mathbb{N}$ satisfy $n \geq 2$. Prove that $a^{1/n} - b^{1/n} < (a-b)^{1/n}$. [*Hint*: Show that $f(x) := x^{1/n} - (x-1)^{1/n}$ is decreasing for $x \geq 1$, and evaluate f at 1 and a/b.]

6. Use the Mean Value Theorem to prove that $|\sin x - \sin y| \leq |x - y|$ for all x, y in $\mathbb{R}$.

7. Use the Mean Value Theorem to prove that $(x-1)/x < \ln x < x - 1$ for $x > 1$. [*Hint*: Use the fact that $D \ln x = 1/x$ for $x > 0$.]

8. Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable in (a, b). Show that if $\lim_{x \to a} f'(x) = A$, then $f'(a)$ exists and equals A. [*Hint*: Use the definition of $f'(a)$ and the Mean Value Theorem.]

9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := 2x^4 + x^4 \sin(1/x)$ for $x \neq 0$ and $f(0) := 0$. Show that f has an absolute minimum at $x = 0$, but that its derivative has both positive and negative values in every neighborhood of 0.

10. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) := x + 2x^2 \sin(1/x)$ for $x \neq 0$ and $g(0) := 0$. Show that $g'(0) = 1$, but in every neighborhood of 0 the derivative $g'(x)$ takes on both positive and negative values. Thus g is not monotonic in any neighborhood of 0.

11. Give an example of a uniformly continuous function on [0,1] that is differentiable on (0, 1) but whose derivative is not bounded on (0, 1).

12. If $h(x) := 0$ for $x < 0$ and $h(x) := 1$ for $x \geq 0$, prove there does not exist a function $f : \mathbb{R} \to \mathbb{R}$ such that $f'(x) = h(x)$ for all $x \in \mathbb{R}$. Give examples of two functions, not differing by a constant, whose derivatives equal $h(x)$ for all $x \neq 0$.

13. Let I be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. Show that if f' is positive on I, then f is strictly increasing on I.

14. Let I be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. Show that if the derivative f' is never 0 on I, then either $f'(x) > 0$ for all $x \in I$ or $f'(x) < 0$ for all $x \in I$.

15. Let I be an interval. Prove that if f is differentiable on I and if the derivative f' is bounded on I, then f satisfies a Lipschitz condition on I. (See Definition 5.4.4.)

16. Let $f : [0, \infty) \to \mathbb{R}$ be differentiable on $(0, \infty)$ and assume that $f'(x) \to b$ as $x \to \infty$.
    (a) Show that for any $h > 0$, we have $\lim_{x\to\infty} (f(x+h) - f(x))/h = b$.
    (b) Show that if $f(x) \to a$ as $x \to \infty$, then $b = 0$.
    (c) Show that $\lim_{x\to\infty} (f(x)/x) = b$.

17. Let f, g be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.

18. Let $I := [a, b]$ and let $f : I \to \mathbb{R}$ be differentiable at $c \in I$. Show that for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - y| < \delta$ and $a \leq x \leq c \leq y \leq b$, then
$$\left| \frac{f(x) - f(y)}{x - y} - f'(c) \right| < \varepsilon.$$

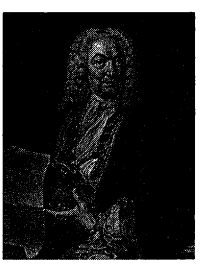
19. A differentiable function $f : I \to \mathbb{R}$ is said to be **uniformly differentiable** on $I := [a, b]$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - y| < \delta$ and $x, y \in I$, then
$$\left| \frac{f(x) - f(y)}{x - y} - f'(x) \right| < \varepsilon.$$
Show that if f is uniformly differentiable on I, then f' is continuous on I.

20. Suppose that $f : [0, 2] \to \mathbb{R}$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$, and that $f(0) = 0, f(1) = 1, f(2) = 1$.
    (a) Show that there exists $c_1 \in (0, 1)$ such that $f'(c_1) = 1$.
    (b) Show that there exists $c_2 \in (1, 2)$ such that $f'(c_2) = 0$.
    (c) Show that there exists $c \in (0, 2)$ such that $f'(c) = 1/3$.

Section 6.3 L'Hospital's Rules

In this section we will discuss limit theorems that involve cases that cannot be determined by previous limit theorems. For example, if $f(x)$ and $g(x)$ both approach 0 as x approaches a, then the quotient $f(x)/g(x)$ may or may not have a limit at a and it is said to have the indeterminate form 0/0. The limit theorem for this case is due to Johann Bernoulli and first appeared in the 1696 book published by L'Hospital.

Johann Bernoulli



Johann Bernoulli (1667–1748) was born in Basel, Switzerland. Johann worked for a year in his father's spice business, but he was not a success. He enrolled in Basel University to study medicine, but his brother Jacob, twelve years older and a Professor of Mathematics, led him into mathematics. Together, they studied the papers of Leibniz on the new subject of calculus. Johann received his doctorate at Basel University and joined the faculty at Groningen in Holland, but upon Jacob's death in 1705, he returned to Basel and was awarded Jacob's chair in mathematics. Because of his many advances in the subject, Johann is regarded as one of the founders of calculus.

While in Paris in 1692, Johann met the Marquis Guillame Francois de L'Hospital and agreed to a financial arrangement under which he would teach the new calculus to L'Hospital, giving L'Hospital the right to use Bernoulli's lessons as he pleased. This was subsequently continued through a series of letters. In 1696, the first book on differential calculus, *L'Analyse des Infiniment Petits*, was published by L'Hospital. Though L'Hospital's name was not on the title page, his portrait was on the frontispiece and the preface states "I am indebted to the clarifications of the brothers Bernoulli, especially the younger." The book contains a theorem on limits later known as L'Hospital's Rule although it was in fact discovered by Johann Bernoulli. In 1922, manuscripts were discovered that confirmed the book consisted mainly of Bernoulli's lessons. And in 1955, the L'Hospital–Bernoulli correspondence was published in Germany.